Data Modeling	2	3
Relational Databases	1	3
Query Construction	2	4
Query Processing		4
DBMS Internals		4
NoSQL Systems		2
Data Security & Privacy	1	2
Data Analytics		3
Distributed Databases/Cloud Computing		
Semi-structured and Unstructured Databases		
Society, Ethics, and the Profession	Included in SEP hours	
Total	**10**	**26**

The CS Core hour in Data Security & Privacy is shared with SEC and is counted here.

Knowledge Units

DM-Data: The Role of Data and the Data Life Cycle

CS Core:

1. The Data Life Cycle: Creation-Processing-Review/Reporting-Retention/Retrieval-Destruction (See also: SEP-Context, SEP-Ethical-Analysis, SEP-Professional-Ethics, SEP-Privacy, SEP-Security, SEC-Foundations)

Illustrative Learning Outcomes:

CS Core:

1. Identify the five stages of the Data Life Cycle.

DM-Core: Core Database System Concepts

CS Core:

1. Purpose and advantages of database systems
2. Components of database systems

3. Design of core DBMS functions (e.g., query mechanisms, transaction management, buffer management, access methods)
4. Database architecture, data independence, and data abstraction
5. Transaction management
6. Normalization
7. Approaches for managing large volumes of data (e.g., NoSQL database systems, use of MapReduce) (See also: PDC-Algorithms)
8. How to support CRUD-only applications
9. Distributed databases/cloud-based systems
10. Structured, semi-structured, and unstructured data
11. Use of a declarative query language

KA Core:

12. Systems supporting structured and/or stream content

Illustrative Learning Outcomes:

CS Core:

1. Identify at least four advantages that using a database system provides.
2. Enumerate the components of a (relational) database system.
3. Follow a query as it is processed by the components of a (relational) database system.
4. Defend the value of data independence.
5. Compose a simple select-project-join query in SQL.
6. Enumerate the four properties of a correct transaction manager.
7. Describe the advantages for eliminating duplicate repeated data.
8. Outline how MapReduce uses parallelism to process data efficiently.
9. Evaluate the differences between structured and semi/unstructured databases.

DM-Modeling: Data Modeling

CS Core:

1. Data modeling (See also: SE-Requirements)
2. Relational data model (See also: MSF-Discrete)

KA Core:

3. Conceptual models (e.g., entity-relationship, UML diagrams)
4. Semi-structured data models (expressed using DTD, XML, or JSON Schema, for example)

Non-core:

5. Spreadsheet models
6. Object-oriented models (See also: FPL-OOP)
    a. GraphQL
7. New features in SQL
8. Specialized Data Modeling topics
    a. Time series data (aggregation, join)
    b. Graph data (link traversal)

c. Techniques for avoiding inefficient raw data access (e.g., “avg daily price”): materialized views and special data structures (e.g., Hyperloglog, bitmap)
d. Geo-Spatial data (e.g., GIS databases) (See also: SPD-Interactive)

Illustrative Learning Outcomes:
CS Core:
1. Describe the components of the relational data model.
2. Model 1:1, 1:n, and n:m relationships using the relational data model.

KA Core:
3. Describe the components of the E-R (or some other non-relational) data model.
4. Model a given environment using a conceptual data model.
5. Model a given environment using the document-based or key-value store-based data model.

DM-Relational: Relational Databases

CS Core:
1. Entity and referential integrity: Candidate key, superkeys
2. Relational database design

KA Core:
3. Mapping conceptual schema to a relational schema
4. Physical database design: file and storage structures (See also: OS-Files)
5. Introduction to Functional dependency theory
6. Normalization Theory
    a. Decomposition of a schema; lossless-join, and dependency-preservation properties of a decomposition
    b. Normal forms (BCNF)
    c. Denormalization (for efficiency)

Non-core:
7. Functional dependency theory
    a. Closure of a set of attributes
    b. Canonical Cover
8. Normalization theory
    a. Multi-valued dependency (4NF)
    b. Join dependency (PJNF, 5NF)
    c. Representation theory

Illustrative Learning Outcomes:
CS Core:
1. Describe the defining characteristics behind the relational data model.
2. Comment on the difference between a foreign key and a superkey.
3. Enumerate the different types of integrity constraints.

KA Core:

4. Compose a relational schema from a conceptual schema which contains 1:1, 1:n, and n:m relationships.
5. Map appropriate file structure to relations and indices.
6. Describe how functional dependency theory generalizes the notion of key.
7. Defend a given decomposition as lossless and or dependency preserving.
8. Detect which normal form a given decomposition yields.
9. Comment on reasons for denormalizing a relation.

DM-Querying: Query Construction

CS Core:

1. SQL Query Formation
    a. Interactive SQL execution
    b. Programmatic execution of an SQL query

KA Core:

2. Relational Algebra
3. SQL
    a. Data definition including integrity and other constraint specifications
    b. Update sublanguage

Non-core:

4. Relational Calculus
5. QBE and 4th-generation environments
6. Different ways to invoke non-procedural queries in conventional languages
7. Introduction to other major query languages (e.g., XPATH, SPARQL)
8. Stored procedures

Illustrative Learning Outcomes:

CS Core:

1. Compose SQL queries that incorporate select, project, join, union, intersection, set difference, and set division.
2. Determine when a nested SQL query is correlated or not.
3. Iterate over data retrieved programmatically from a database via an SQL query.

KA Core:

4. Define, in SQL, a relation schema, including all integrity constraints and delete/update triggers.
5. Compose an SQL query to update a tuple in a relation.

DM-Processing: Query Processing

KA Core:

1. Page structures
2. Index structures
    a. B+ trees (See also: AL-Foundational)

    b. Hash indices: static and dynamic (See also: AL-Foundational, SEC-Foundations)
    c. Index creation in SQL
3. File structures (See also: OS-Files)
    a. Heap files
    b. Hash files
4. Algorithms for query operators
    a. External Sorting (See also: AL-Foundational)
    b. Selection
    c. Projection; with and without duplicate elimination
    d. Natural Joins: Nested loop, Sort-merge, Hash join
    e. Analysis of algorithm efficiency (See also: AL-Complexity)
5. Query transformations
6. Query optimization
    a. Access paths
    b. Query plan construction
    c. Selectivity estimation
    d. Index-only plans
7. Parallel Query Processing (e.g., parallel scan, parallel join, parallel aggregation) (See also: PDC-Algorithms)
8. Database tuning/performance
    a. Index selection
    b. Impact of indices on query performance (See also: SF-Performance, SEP-Sustainability)
    c. Denormalization

Illustrative Learning Outcomes:

KA Core:

1. Describe the purpose and organization of both B+ tree and hash index structures.
2. Compose an SQL command to create an index (any kind).
3. Specify the steps for the various query operator algorithms: external sorting, projection with duplicate elimination, sort-merge join, hash-join, block nested-loop join.
4. Derive the run-time (in I/O requests) for each of the above algorithms.
5. Transform a query in relational algebra to its equivalent appropriate for a left-deep, pipelined execution.
6. Compute selectivity estimates for a given selection and/or join operation.
7. Describe how to modify an index structure to facilitate an index-only operation for a given relation.
8. For a given scenario decide on which indices to support for the efficient execution of a set of queries.
9. Describe how DBMSs leverage parallelism to speed up query processing by dividing the work across multiple processors or nodes.

DM-Internals: DBMS Internals

KA Core:

1. DB Buffer Management (See also: OS-Memory, SF-Resource)

2. Transaction Management (See also: PDC-Coordination)
    a. Isolation Levels
    b. ACID
    c. Serializability
    d. Distributed Transactions
3. Concurrency Control: (See also: OS-Concurrency)
    a. 2-Phase Locking
    b. Deadlocks handling strategies
    c. Quorum-based consistency models
4. Recovery Manager
    a. Relation with Buffer Manager

Non-core:

5. Concurrency Control:
    a. Optimistic concurrency control
    b. Timestamp concurrency control
6. Recovery Manager
    a. Write-Ahead logging
    b. ARIES recovery system (Analysis, REDO, UNDO)

Illustrative Learning Outcomes:

KA Core:

1. Describe how a DBMS manages its Buffer Pool.
2. Describe the four properties for a correct transaction manager.
3. Outline the principle of serializability.

DM-NoSQL: NoSQL Systems

KA Core:

1. Why NoSQL? (e.g., Impedance mismatch between Application [CRUD] and RDBMS)
2. Key-Value and Document data model

Non-core:

3. Storage systems (e.g., Key-Value systems, Data Lakes)
4. Distribution Models (Sharding and Replication) (See also: PDC-Communication)
5. Graph Databases
6. Consistency Models (Update and Read, Quorum consistency, CAP theorem) (See also: PDC-Communication)
7. Processing model (e.g., Map-Reduce, multi-stage map-reduce, incremental map-reduce) (See also: PDC-Communication)
8. Case Studies: Cloud storage system (e.g., S3); Graph databases; “When not to use NoSQL” (See also: SPD-Web)

Illustrative Learning Outcomes:

KA Core:

1. Develop a use case for the use of NoSQL over RDBMS.
2. Describe the defining characteristics behind Key-Value and Document-based data models.

DM-Security: Data Security and Privacy

CS Core:

1. Differences between data security and data privacy (See also: SEC-Foundations)
2. Protecting data and database systems from attacks, including injection attacks such as SQL injection (See also: SEC-Foundations)
3. Personally identifying information (PII) and its protection (See also: SEC-Foundations, SEP-Security, SEP-Privacy)
4. Ethical considerations in ensuring the security and privacy of data (See also: SEC-SEP, SEP-Ethical-Analysis, SEP-Security, SEP-Privacy)

KA Core:

5. Need for, and different approaches to securing data at rest, in transit, and during processing (See also: SEC-Foundations, SEC-Crypto)
6. Database auditing and its role in digital forensics (See also: SEC-Forensics)
7. Data inferencing and preventing attacks (See also: SEC-Crypto)
8. Laws and regulations governing data security and data privacy (See also: SEP-Security, SEP-Privacy, SEC-Foundations, SEC-Governance)

Non-core:

9. Typical risk factors and prevention measures for ensuring data integrity (See also: SEC-Governance)
10. Ransomware and prevention of data loss and destruction (See also: SEC-Coding, SEC-Forensics)

Illustrative Learning Outcomes:

CS Core:

1. Describe the differences in the goals for data security and data privacy.
2. Identify and mitigate risks associated with different approaches to protecting data.
3. Describe legal and ethical considerations of end-to-end data security and privacy.

KA Core:

4. Develop a database auditing system given risk considerations.
5. Apply several data exploration approaches to understanding unfamiliar datasets.

DM-Analytics: Data Analytics

KA Core:

1. Exploratory data techniques (motivation, representation, descriptive statistics, visualizations)
2. Data science lifecycle: business understanding, data understanding, data preparation, modeling, evaluation, deployment, and user acceptance (See also: AI-ML)

3. Data mining and machine learning algorithms: e.g., classification, clustering, association, regression (See also: AI-ML)
4. Data acquisition and governance (See also: SEC-Governance)
5. Data security and privacy considerations (See also: SEP-Security, SEP-Privacy, SEC-Foundations)
6. Data fairness and bias (See also: SEP-Security, AI-SEP)
7. Data visualization techniques and their use in data analytics (See also: GIT-Visualization)
8. Entity Resolution

Illustrative Learning Outcomes:

KA Core:

1. Describe several data exploration approaches, including visualization, to understanding unfamiliar datasets.
2. Apply several data exploration approaches to understanding unfamiliar datasets.
3. Describe basic machine learning/data mining algorithms and when they are appropriate for use.
4. Apply several machine learning/data mining algorithms.
5. Describe legal and ethical considerations in acquiring, using, and modifying datasets.
6. Describe issues of fairness and bias in data collection and usage.

DM-Distributed: Distributed Databases/Cloud Computing

Non-core:

1. Distributed DBMS (See also: PDC-Communications)
    a. Distributed data storage
    b. Distributed query processing
    c. Distributed transaction model
    d. Homogeneous and heterogeneous solutions
    e. Client-server distributed databases (See also: NC-Fundamentals)
2. Parallel DBMS (See also: PDC-Algorithms)
    a. Parallel DBMS architectures: shared memory, shared disk, shared nothing;
    b. Speedup and scale-up, e.g., use of the MapReduce processing model (See also: PDC-Programs, SF-Foundations)
    c. Data replication and weak consistency models (See also: PDC-Coordination)

DM-Unstructured: Semi-structured and Unstructured Databases

Non-core:

1. Vectorized unstructured data (text, video, audio, etc.) and vector storage
    a. TF-IDF Vectorizer with ngram
    b. Word2Vec
    c. Array database or array data type handling
2. Semi-structured databases (e.g., JSON)
    a. Storage
        i. Encoding and compression of nested data types
    b. Indexing
        i. Btree, skip index, Bloom filter

ii. Inverted index and bitmap compression
iii. Space filling curve indexing for semi-structured geo-data

c. Query processing for OLTP and OLAP use cases
i. Insert, Select, update/delete tradeoffs
ii. Case studies on Postgres/JSON, MongoDB, and Snowflake/JSON

DM-SEP: Society, Ethics, and the Profession

CS Core:

1. Issues related to scale (See also: SEP-Economies)
2. Data privacy overall (See also: SEP-Privacy, SEP-Ethical-Analysis)
    a. Privacy compliance by design (See also: SEP-Privacy)
3. Data anonymity (See also: SEP-Privacy)
4. Data ownership/custodianship (See also: SEP-Professional-Ethics)
5. Intended and unintended applications of stored data (See also: SEP-Professional-Ethics, SEC-Foundations)

KA Core:

6. Reliability of data (See also: SEP-Security)
7. Provenance, data lineage, and metadata management (See also: SEP-Professional-Ethics)
8. Data security (See also: DM-Security, SEP-Security)

Illustrative Learning Outcomes:

CS Core:

1. Enumerate three social and three legal issues related to large data collections.
2. Describe the value of data privacy.
3. Identify the competing stakeholders with respect to data ownership.
4. Enumerate three negative unintended consequences from a given (well known) data-centric application (e.g., Facebook, LastPass, Ashley Madison).

KA Core:

5. Describe the meaning of data provenance and lineage.
6. Identify how a database might contribute to data security as well as how it may introduce insecurities.

Professional Dispositions

- **Meticulous:** Those who either access or store data collections must be meticulous in fulfilling data ownership responsibilities.
- **Responsible**: In conjunction with the professional management of (personal) data, it is equally important that data are managed responsibly. Protection from unauthorized access as well as prevention of irresponsible, though legal, use of data is paramount. Furthermore, data custodians need to protect data not only from outside attack, but from crashes and other foreseeable dangers.